



Facultad de
Ciencias
UNAM

Characterization of plasmonic disordered metasurfaces for biosensing applications under realistic conditions

Examen de candidatura

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Departamento de Física

Facultad de Ciencias

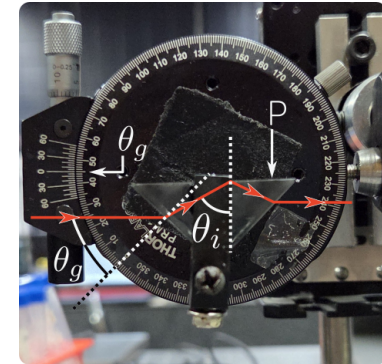
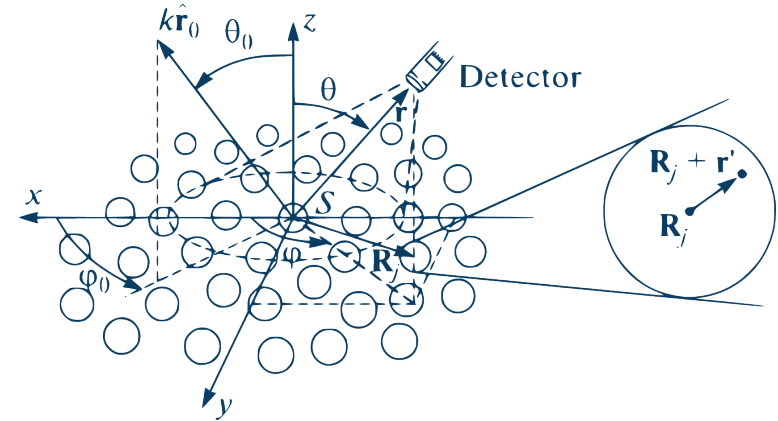
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Contenido

1. Antecedentes y objetivos del proyecto
2. Teorías de esparcimiento múltiple y técnicas de medición
 - 2.1. Modelo de esparcimiento coherente
 - 2.2. Teoría de película delgada
 - 2.3. Arreglos experimentales
3. Avances en caracterización no intrusiva
 - 3.1. Efectos debido al sustrato
4. Avances en respuesta óptica y mecanismos de sensado de metasuperficies
 - 4.1. Diseño y uso de oscuridad topológica
5. Publicaciones y manuscritos
6. Plan de trabajo 2025 - 2027



Antecedentes

Metasuperficies para biosensado

Metasuperficie plasmónica: Arreglo bidimensional de nanoestructuras metálicas soportadas por un sustrato.¹

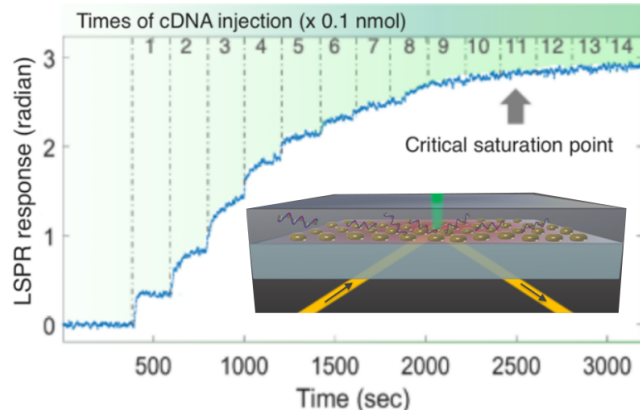
Sintonización de propiedades ópticas:

- Geometría
- Material
- Distribución espacial

¹A. K. González-Alcalde et al.
Opt Commun, **475**:126289, 2020

Esquemas de medición:

- Reflexión²
- Elipsometría^{3,4}



³G. Qiu et al. ACS Nano, **14**(5):5268–5277, 2020

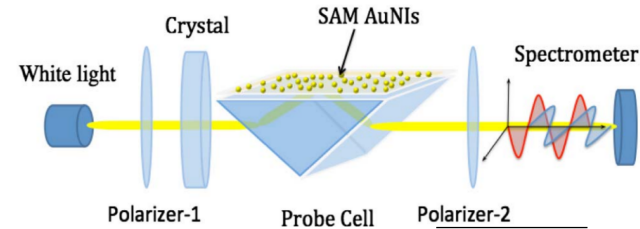
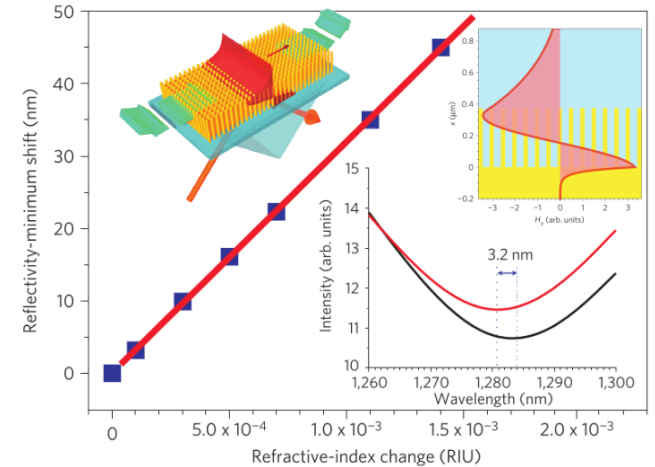
Dewetting

$d = 5$ nm
(Grosor película)

²A. V. Kabashin et al. Nat Mater, **8**(11):867–871, 2009

Crecimiento electroquímico
Au/Al₂O₃

$L = 380$ nm
 $a = 12.5$ nm
 $\Lambda = 60$ nm



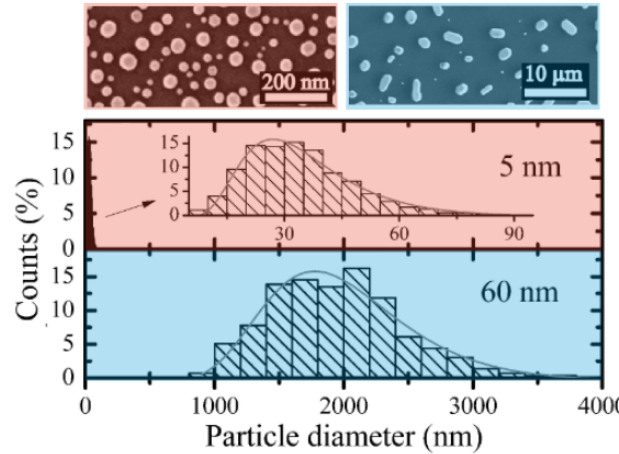
⁴G. Qiu et al. Opt. Lett., **40**(9):1924, 2015

Metasuperficies en condiciones realistas

Polidispersidad y efectos de sustrato

Polidispersidad de tamaños

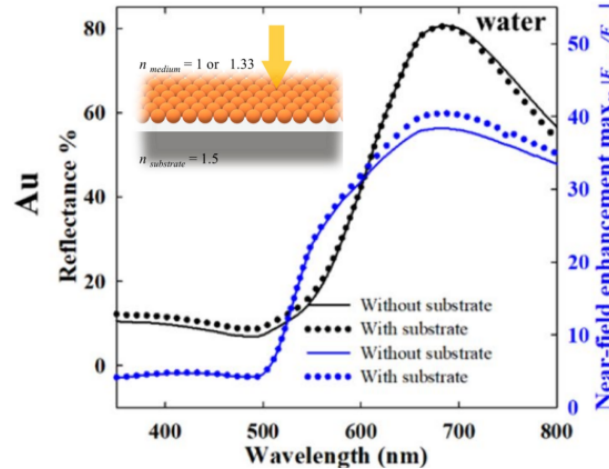
Enfoque analítico



D. Wang et al. Beilstein J. Nanotechnol., **2**:318–326, 2011

Presencia del sustrato

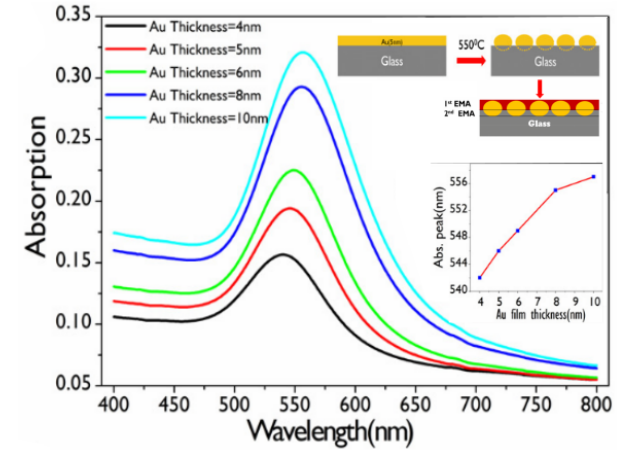
Enfoque analítico
Cálculo numérico



R. Borah et al. Sci Rep, **12**(1):15738, 2022

Incrustamiento

Enfoque analítico
Cálculo numérico



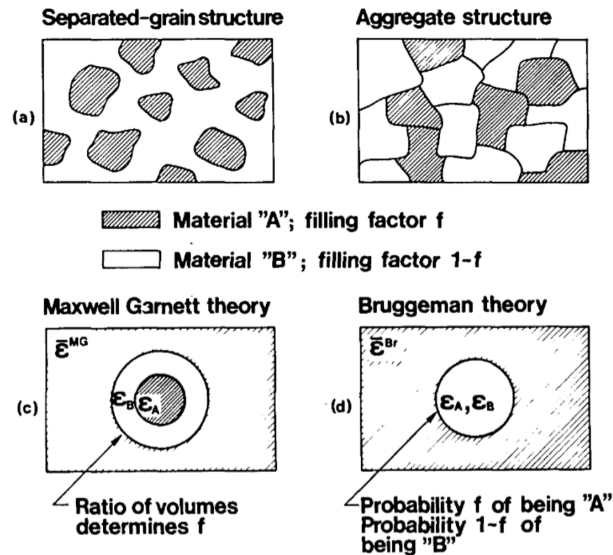
R. S. Moirangthem et al. Biomed. Opt. Express, **3**(5):899, 2012

Respuesta óptica de metasuperficies

Enfoques analíticos

Teorías de medio efectivo¹

- Reemplazo de medio heterogéneo por **homogéneo**
- **Ejemplos:** Maxwell Garnett y Bruggeman
- Usualmente para sistemas coloidales
- **Correcciones** fenomenológicas para uso en metasuperficies²

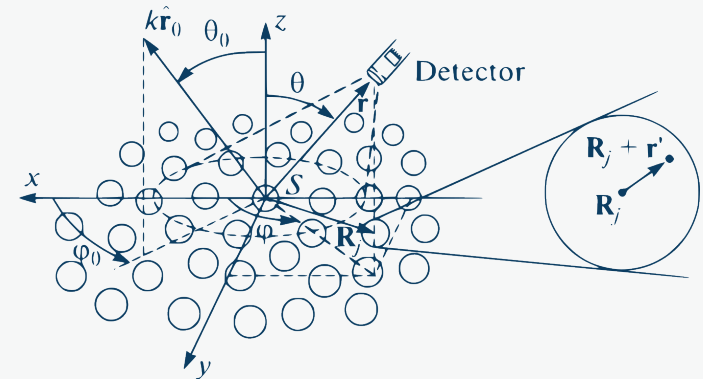


¹G. A. Niklasson et al. Appl. Opt., **20**(1):26, 1981

²A. V. Kabashin et al. Nat Mater, **8**(11):867-871, 2009

Teorías de esparcimiento múltiple³

- **Ensamble de partículas** interactuantes
- Solución aproximada a las ecuaciones de Maxwell
- **Ejemplos:** Modelo de Esparcimiento Coherente⁴ (CSM)
Teoría de Película Delgada⁵ (TFT)
- Varios órdenes de aproximación:
 - En el orden del esparcimiento múltiple
 - En el orden multipolar de cada esparcador



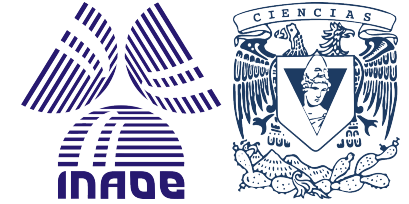
³N. A. Loiko et al. J. Exp. Theor. Phys., **131**(2):227-243, 2020

⁴A. Reyes-Coronado et al. Optics Express, **9594**:6697-6706, 2018

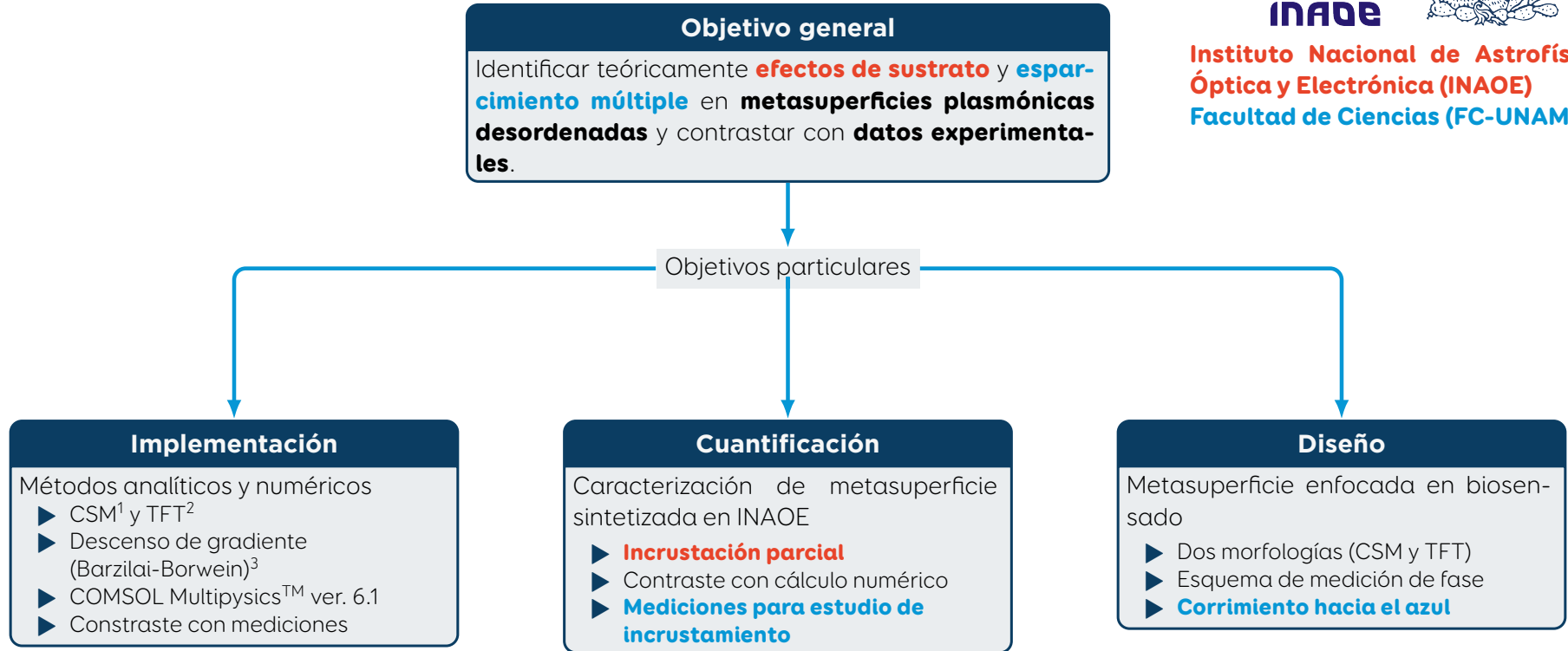
⁵D. Bedeaux et al. Optical properties of surfaces. Imperial College Press, London, 2.^a edición, 2004

Objetivos

Colaboración teórico-experimental



**Instituto Nacional de Astrofísica,
Óptica y Electrónica (INAOE)**
Facultad de Ciencias (FC-UNAM)



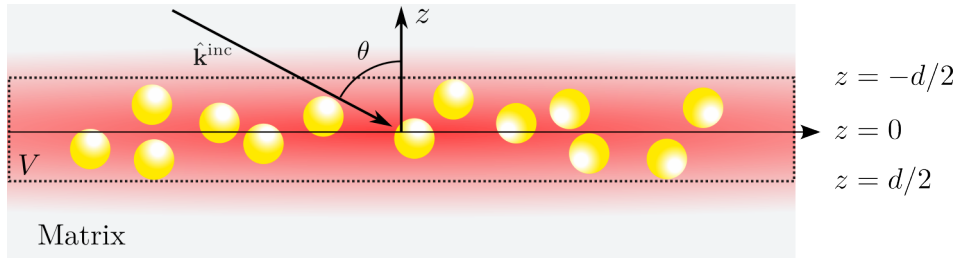
¹A. Reyes-Coronado et al. Optics Express, **9594**:6697-6706, 2018
²D. Bedeaux et al. Optical properties of surfaces. Imperial College Press, London, 2.^a edición, 2004
³J. Barzilai et al. IMA J Numer Anal, **8**(1):141-148, 1988

Teoría de esparcimiento múltiple¹

Jerarquización de Foldy-Lax

N esparcidores

- Iluminados con onda plana
- $N \rightarrow \infty$ ecuaciones
- V : Volumen semiinfinito
- Promedio configuracional: $\rho(\mathbf{R})$



$$\mathbf{E}_k^{\text{exc}}(\mathbf{r}) = \mathbf{E}^{\text{inc}}(\mathbf{r}) + \sum_{\ell \neq k}^N \mathbf{E}_\ell^{\text{ind}}(\mathbf{r})$$

$$\mathbf{E}_\ell^{\text{ind}}(\mathbf{r}) = \int d^3 r' \mathbb{G}(\mathbf{r}, \mathbf{r}') \times \int d^3 r'' \mathbb{T}(\mathbf{r}' - \mathbf{r}_\ell, \mathbf{r}'' - \mathbf{r}_\ell) \mathbf{E}_\ell^{\text{exc}}(\mathbf{r}'')$$

$$\langle \mathbf{E}(\mathbf{r}) \rangle = \mathbf{E}^{\text{inc}}(\mathbf{r}) + \sum_{\ell=1}^N \underbrace{\left(\prod_{k=1}^N \int d^3 r_k \rho(\mathbf{R}) \mathbf{E}_\ell^{\text{ind}}(\mathbf{r}) \right)}_{\langle \mathbf{E}_\ell^{\text{ind}} \rangle}$$

► Aproximación de esparcidor individual (SSA)

$$\langle \mathbf{E}_\ell^{\text{exc}}(\mathbf{r}'', \mathbf{R}) \rangle_\ell = \mathbf{E}^{\text{inc}}(\mathbf{r}'')$$

► Aproximación cuasicristalina

$$\langle \mathbf{E}_m^{\text{exc}}(\mathbf{r}'', \mathbf{R}) \rangle_{\ell, m} = \langle \mathbf{E}_\ell^{\text{exc}}(\mathbf{r}'', \mathbf{R}) \rangle_\ell$$

¹A. García-Valenzuela et al. JOSA A, **29**(6):1161-1179, 2012

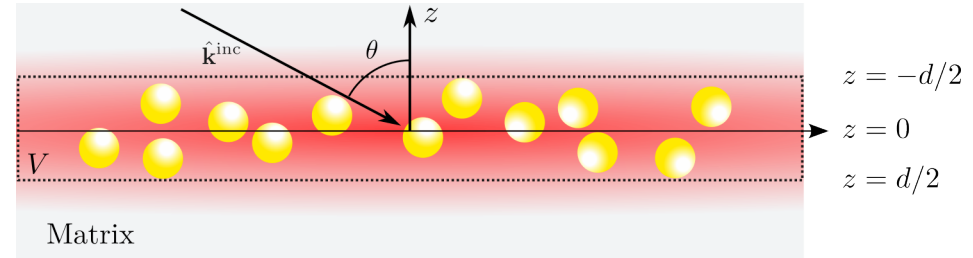
Modelo de esparcimiento coherente (CSM)

Caso monodisperso¹

N esparcidores

- **QCA**
- Película de grosor $d \rightarrow 0$
- Correlación de esfera dura
- Esferas de radio a
- $\rho(\mathbf{r}_\ell) = 1/V$
- $\alpha = 2\Theta/[(ka)^2 \cos \theta]$

$$\langle \mathbf{E}_{\text{QCA}}^{\text{ind}}(\mathbf{r}) \rangle = \begin{cases} -\alpha \left(S_j(0) \|\mathbf{E}_r^{\text{exc}}(\mathbf{r})\| + S(\pi - 2\theta) \|\mathbf{E}_t^{\text{exc}}(\mathbf{r})\| \right) \exp(i\mathbf{k}_{\text{coh}}^r \cdot \mathbf{r}) \hat{\mathbf{E}}^{\text{inc}}, & z > 0 \\ -\alpha \left(S_j(\pi - 2\theta) \|\mathbf{E}_r^{\text{exc}}(\mathbf{r})\| + S(0) \|\mathbf{E}_t^{\text{exc}}(\mathbf{r})\| \right) \exp(i\mathbf{k}_{\text{coh}}^t \cdot \mathbf{r}) \hat{\mathbf{E}}^{\text{inc}}, & z < 0 \end{cases}$$



Coefficiente de reflexión y transmisión

$$r_{\text{CSM}}^{(j)}(\theta) = \frac{-\alpha S_j(\pi - 2\theta)}{1 + \alpha S(0) + \frac{1}{4}\alpha^2 [S^2(0) - S_j^2(\pi - 2\theta)]}$$

$$t_{\text{CSM}}^{(j)}(\theta) = \frac{1 - \frac{1}{4}\alpha^2 [S^2(0) - S_j^2(\pi - 2\theta)]}{1 + \alpha S(0) + \frac{1}{4}\alpha^2 [S^2(0) - S_j^2(\pi - 2\theta)]}$$

Múltiples reflexiones

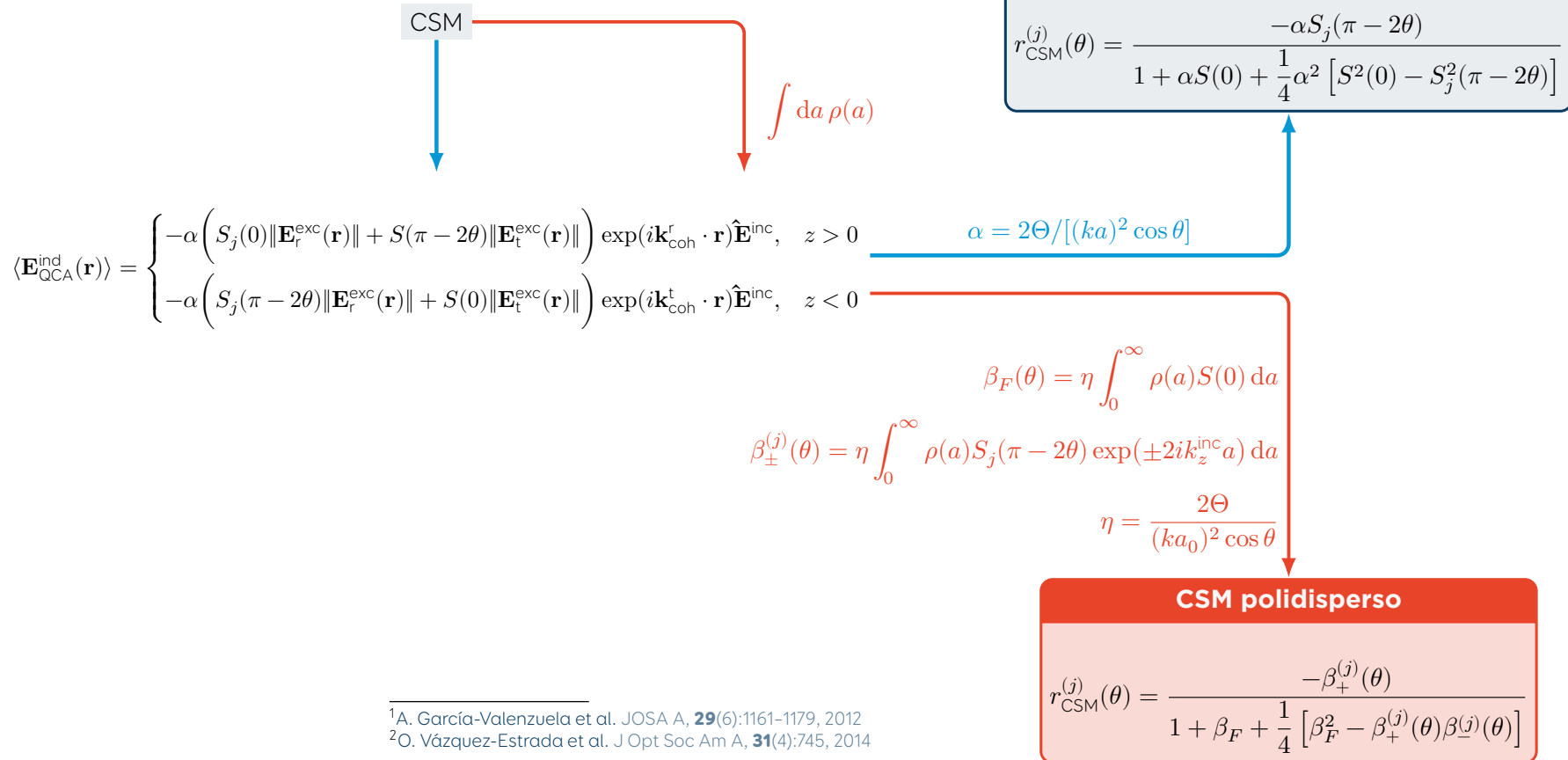
Presencia de sustrato (Reflexión)

$$r^{(j)}(\theta_i) = \frac{r_{\text{sm}}^{(j)}(\theta_i) + r_{\text{CSM}}^{(j)}(\theta_t)}{1 + r_{\text{sm}}^{(j)}(\theta_i) r_{\text{CSM}}^{(j)}(\theta_t)}$$

¹A. Reyes-Coronado et al. Optics Express, **9594**:6697–6706, 2018

Modelo de esparcimiento coherente (CSM)

Caso monodisperso¹ y caso polidisperso²



Campos, corrientes y cargas de exceso¹

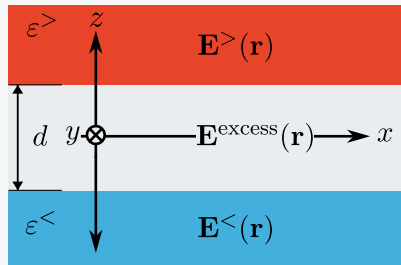
Contribución total del exceso

Ecuaciones de Maxwell (Campos de exceso)

Sistema estratificado

$$\mathbf{E}(\mathbf{r}) = \mathbf{E}^>(\mathbf{r})\Theta(z) + \mathbf{E}^<(\mathbf{r})\Theta(-z) + \mathbf{E}^{\text{excess}}(\mathbf{r})$$

$$\mathbf{E}(\mathbf{r}; z \rightarrow \pm\infty) \rightarrow \mathbf{E}^{\geq}(\mathbf{r})$$



- Tres medios semiinfinitos
- $\epsilon^{\geq}$ para medios en extremos
- $\mathbf{E}^{\geq}(\mathbf{r})$: Campo radiativo
- Descomposición válida para
 - $\mathbf{B}(\mathbf{r})$ y $\mathbf{H}(\mathbf{r})$
 - $\mathbf{D}(\mathbf{r})$
 - $\mathbf{J}_{\text{ext}}(\mathbf{r})$ y $\rho_{\text{ext}}(\mathbf{r})$

$$\nabla \cdot \mathbf{D}^{\text{excess}}(\mathbf{r}) + [D_{\perp}^>(\mathbf{r}_{\parallel}) - D_{\perp}^<(\mathbf{r}_{\parallel})]\delta(z) = \rho_{\text{ext}}^{\text{excess}}(\mathbf{r})$$

$$\nabla \cdot \mathbf{B}^{\text{excess}}(\mathbf{r}) + [B_{\perp}^>(\mathbf{r}_{\parallel}) - B_{\perp}^<(\mathbf{r}_{\parallel})]\delta(z) = 0$$

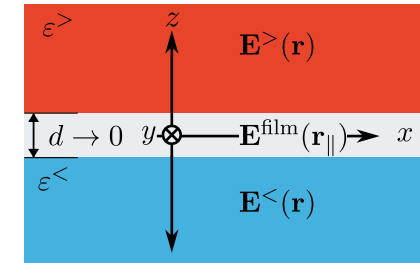
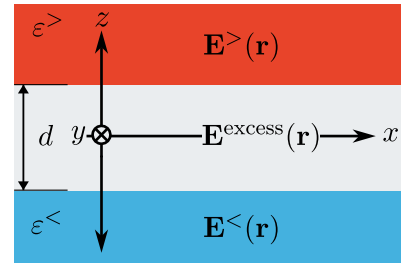
$$\nabla \times \mathbf{E}^{\text{excess}}(\mathbf{r}) + \hat{\mathbf{e}}_{\perp} \times [\mathbf{E}_{\parallel}^>(\mathbf{r}_{\parallel}) - \mathbf{E}_{\parallel}^<(\mathbf{r}_{\parallel})]\delta(z) = +i\omega\mathbf{B}^{\text{excess}}(\mathbf{r})$$

$$\nabla \times \mathbf{H}^{\text{excess}}(\mathbf{r}) + \hat{\mathbf{e}}_{\perp} \times [\mathbf{H}_{\parallel}^>(\mathbf{r}_{\parallel}) - \mathbf{H}_{\parallel}^<(\mathbf{r}_{\parallel})]\delta(z) = \mathbf{J}_{\text{ext}}^{\text{excess}}(\mathbf{r}) - i\omega\mathbf{D}^{\text{excess}}(\mathbf{r})$$

Límite de película delgada

$$\mathbf{E}(\mathbf{r}) = \mathbf{E}^>(\mathbf{r})\Theta(z) + \mathbf{E}^<(\mathbf{r})\Theta(-z) + \mathbf{E}^{\text{film}}(\mathbf{r}_{\parallel})\delta(z)$$

$$\mathbf{E}^{\text{film}}(\mathbf{r}_{\parallel}) = \int_{-\infty}^{\infty} \mathbf{E}^{\text{excess}}(\mathbf{r}) dz = \left(\int_{-\infty}^{\infty} \mathbf{E}^{\text{excess}}(\mathbf{r}) \exp(ik_z z) dz \right) \Big|_{k_z=0}$$



¹D. Bedeaux et al. Optical properties of surfaces. Imperial College Press, London, 2.^a edición, 2004

Teoría de película delgada (TFT)

Condiciones de frontera modificadas¹

Ecuaciones de Maxwell para campos de exceso

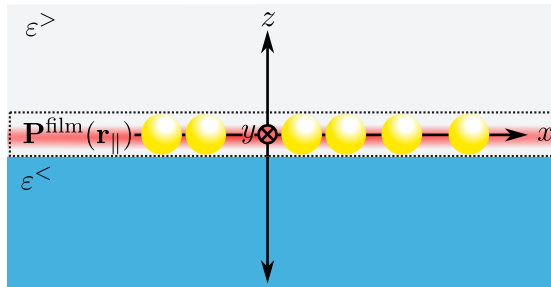
Límite de película delgada
Transformada de Fourier en z
 $k_z = 0$

$$\langle a^{\text{film}}(\mathbf{r}_{\parallel}) \rangle = \frac{1}{2} [a^>(\mathbf{r}_{\parallel}) + a^<(\mathbf{r}_{\parallel})]$$

Ecuaciones constitutivas

$$\mathbf{P}^{\text{film}}(\mathbf{r}_{\parallel}) = \gamma \langle \mathbf{E}_{\parallel}^{\text{film}}(\mathbf{r}_{\parallel}) \rangle + \beta \langle D_{\perp}^{\text{film}}(\mathbf{r}_{\parallel}) \rangle \hat{\mathbf{e}}_{\perp}$$

$$\mathbf{M}^{\text{film}}(\mathbf{r}_{\parallel}) = \mathbf{0}$$



$$\begin{aligned} \blacktriangleright \gamma &= \frac{2}{3} \frac{\Theta}{V_{\text{sp}}} \left(\frac{\alpha_{\text{sp}}}{1 + A\alpha_{\text{sp}}} \right) \\ \blacktriangleright \beta &= \frac{2}{3} \frac{\Theta}{\varepsilon_i^2 V_{\text{sp}}} \left(\frac{\alpha_{\text{sp}}}{1 + 2A\alpha_{\text{sp}}} \right) \\ \blacktriangleright \alpha_{\text{sp}} &= 3\varepsilon_t V_{\text{sp}} \left(\frac{\varepsilon_{\text{sp}} - \varepsilon_t}{\varepsilon_{\text{sp}} + 2\varepsilon_t} \right) \\ \blacktriangleright A &= \frac{\varepsilon_i - \varepsilon_t}{\varepsilon_i + \varepsilon_t} \end{aligned}$$

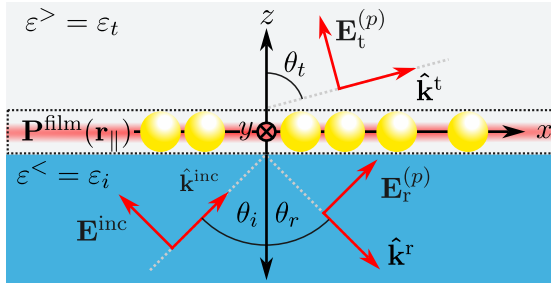
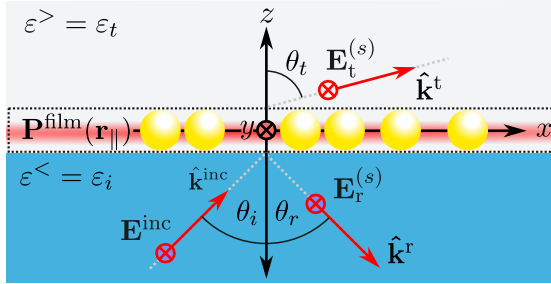
Condiciones de frontera

$$\begin{aligned} [D_{\perp}^>(\mathbf{r}_{\parallel}) - D_{\perp}^<(\mathbf{r}_{\parallel})] &= -\nabla_{\parallel} \cdot \mathbf{P}_{\parallel}^{\text{film}}(\mathbf{r}_{\parallel}) \\ [B_{\perp}^>(\mathbf{r}_{\parallel}) - B_{\perp}^<(\mathbf{r}_{\parallel})] &= -\nabla_{\parallel} \cdot \mathbf{M}_{\parallel}^{\text{film}}(\mathbf{r}_{\parallel}) \\ [\mathbf{E}_{\parallel}^>(\mathbf{r}_{\parallel}) - \mathbf{E}_{\parallel}^<(\mathbf{r}_{\parallel})] &= -i\omega \hat{\mathbf{e}}_{\perp} \times \mathbf{M}_{\parallel}^{\text{film}}(\mathbf{r}_{\parallel}) - \nabla_{\parallel} P_{\perp}^{\text{film}}(\mathbf{r}_{\parallel}) \\ [\mathbf{H}_{\parallel}^>(\mathbf{r}_{\parallel}) - \mathbf{H}_{\parallel}^<(\mathbf{r}_{\parallel})] &= +i\omega \hat{\mathbf{e}}_{\perp} \times \mathbf{P}_{\parallel}^{\text{film}}(\mathbf{r}_{\parallel}) - \nabla_{\parallel} M_{\perp}^{\text{film}}(\mathbf{r}_{\parallel}) \end{aligned}$$

¹D. Bedeaux et al. Optical properties of surfaces. Imperial College Press, London, 2.^a edición, 2004

Teoría de película delgada (TFT)

Coeficientes de Fresnel modificado¹



Polarización: *s*

$$r_{\text{TFT}}^{(s)}(\theta_i) = \frac{n_i \cos \theta_i - n_t \cos \theta_t + i(\omega/c)\gamma}{n_i \cos \theta_i + n_t \cos \theta_t - i(\omega/c)\gamma}$$

$$t_{\text{TFT}}^{(s)}(\theta_i) = \frac{2n_i \cos \theta_i}{n_i \cos \theta_i + n_t \cos \theta_t - i(\omega/c)\gamma},$$

Polarización: *p*

$$r_{\text{TFT}}^{(p)}(\theta_i) = \frac{\kappa_-(\omega) - i(\omega/c)\gamma \cos \theta_i \cos \theta_t + i(\omega/c)n_i n_t \varepsilon_i \beta \sin^2 \theta_i}{\kappa_+(\omega) - i(\omega/c)\gamma \cos \theta_i \cos \theta_t - i(\omega/c)n_i n_t \varepsilon_i \beta \sin^2 \theta_i}$$

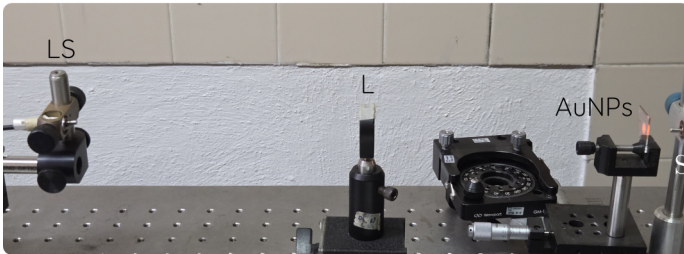
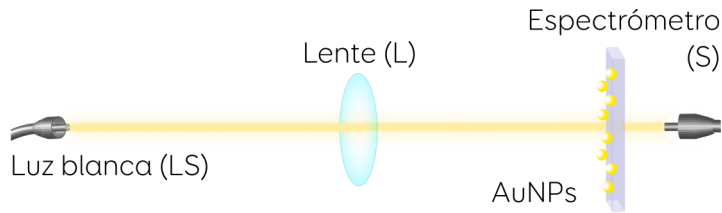
$$t_{\text{TFT}}^{(p)}(\theta_i) = \frac{n_i \cos \theta_i (1 - (\omega/2c)^2 \varepsilon_i \gamma \beta \sin^2 \theta_i)}{\kappa_+(\omega) - i(\omega/c)\gamma \cos \theta_i \cos \theta_t - i(\omega/c)n_i n_t \varepsilon_i \beta \sin^2 \theta_i}$$

$$\kappa_{\pm}(\omega) = \left(n_t \cos \theta_i \pm n_i \cos \theta_t \right) \left[1 - \left(\frac{\omega}{2c} \right)^2 \varepsilon_i \gamma \beta \sin^2 \theta_i \right]$$

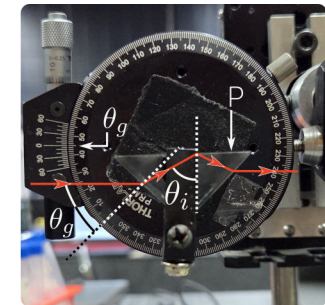
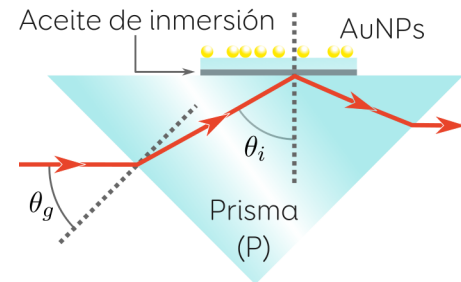
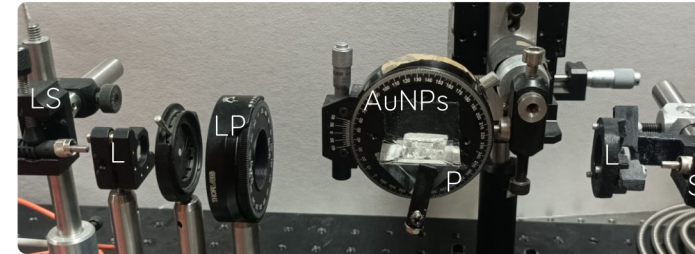
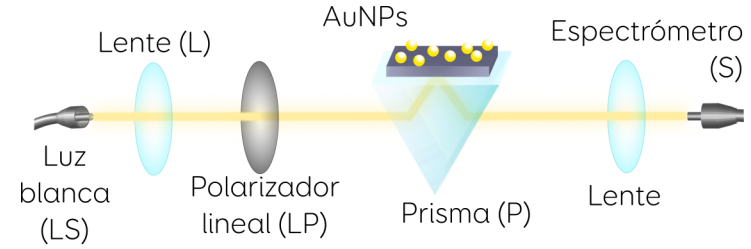
¹D. Bedeaux et al. Optical properties of surfaces. Imperial College Press, London, 2.^a edición, 2004

Arreglos experimentales

Espectro de extinción



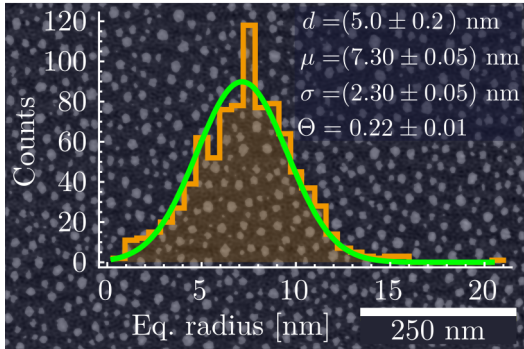
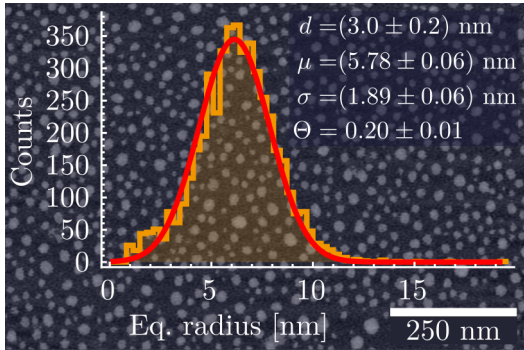
Espectro de reflexión



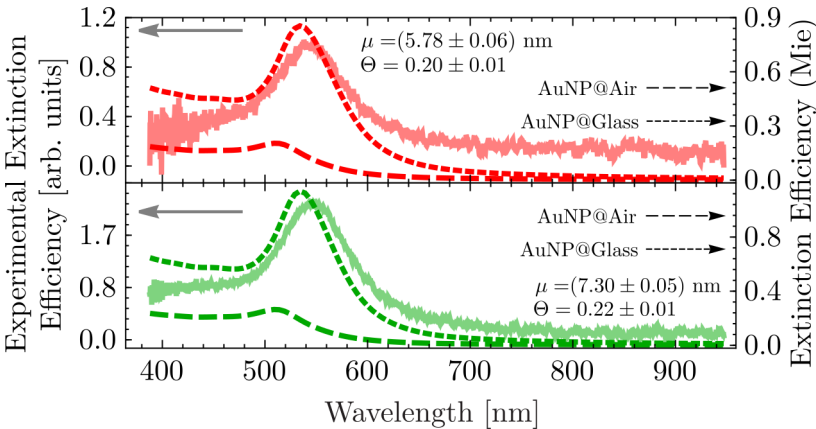
Arreglos montandos en el INAOE por el Dr.
Juan Pablo Cuanalo.

Avances en caracterización no intrusiva

¿Esparcimiento múltiple o interacción con partícula imagen?

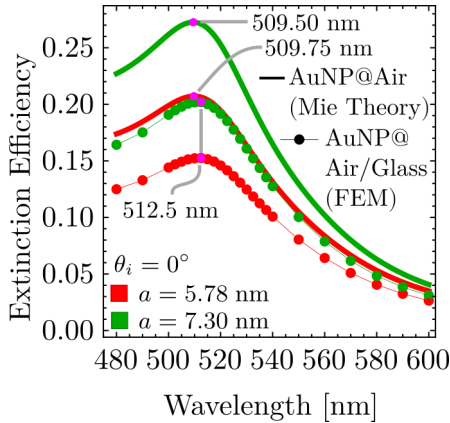


Metasuperficie vs. partícula individual



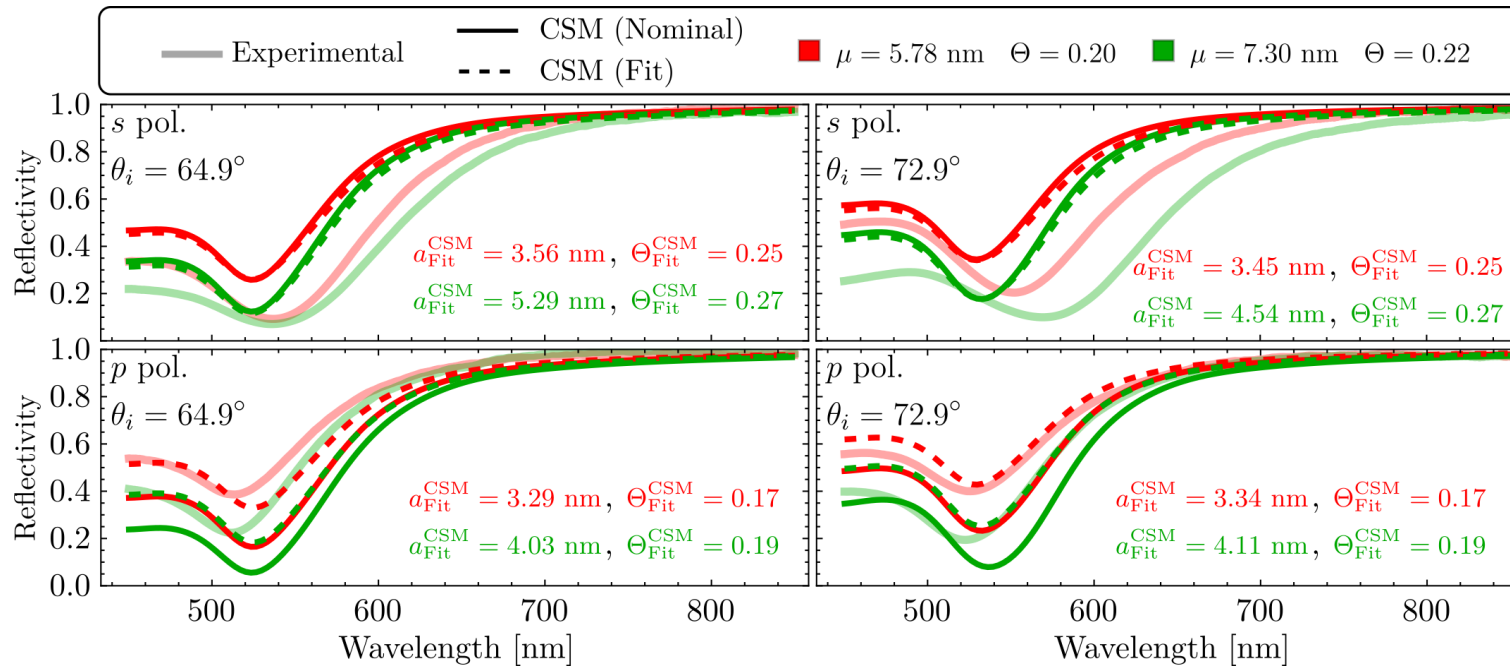
Radio $a = \mu$	5.78 nm	7.30 nm
Metasuperficie (Experimental)	547.19 nm	544.09 nm
AuNP @ Air (Teoría de Mie)	509.75 nm	509.50 nm
AuNP @ Glass (Teoría de Mie)	533.25 nm	533.50 nm

Método de elemento finito: Introduciendo el sustrato



Avances en caracterización no intrusiva

Minimización del error por descenso de gradiente

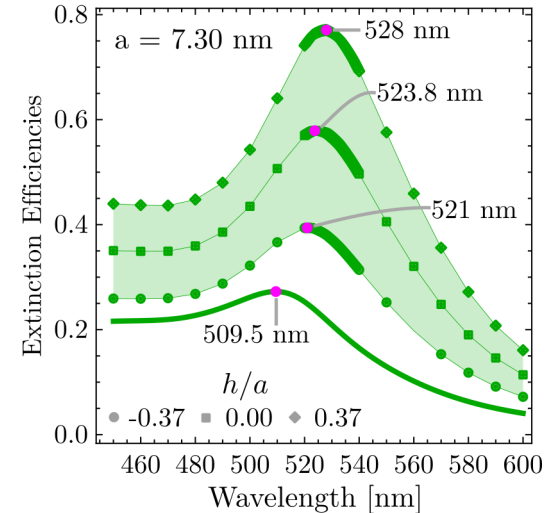
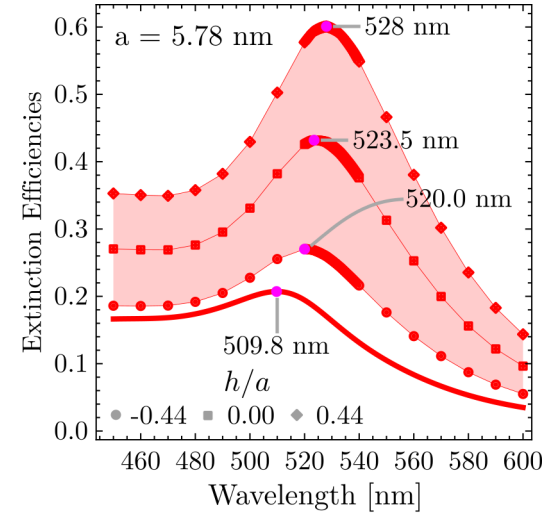
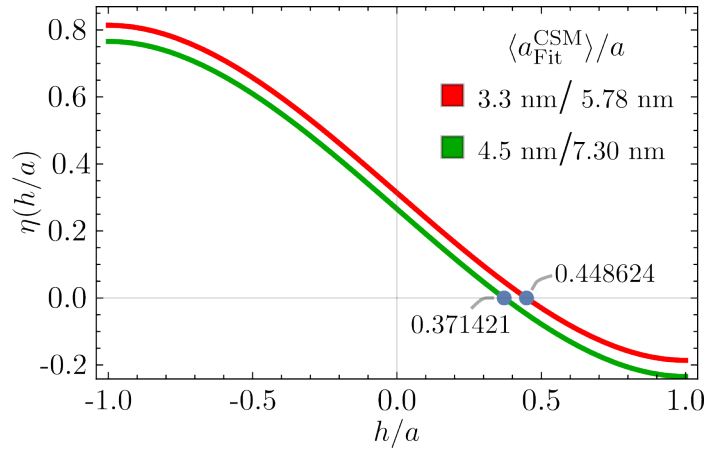
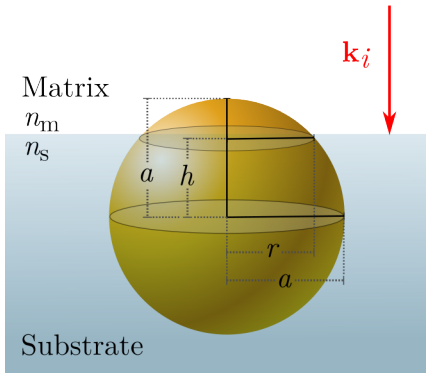


Avances en caracterización no intrusiva

Determinación del incrustamiento parcial

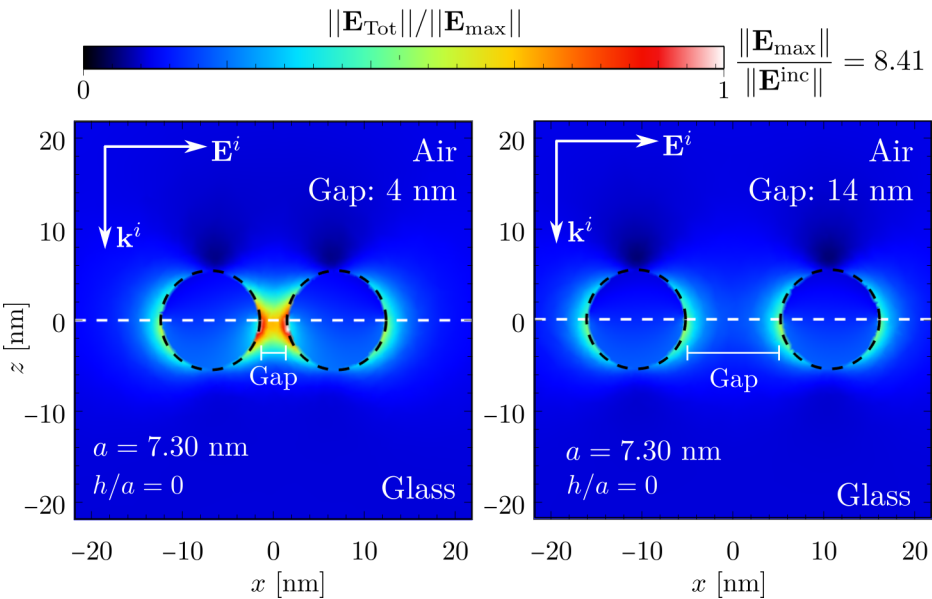
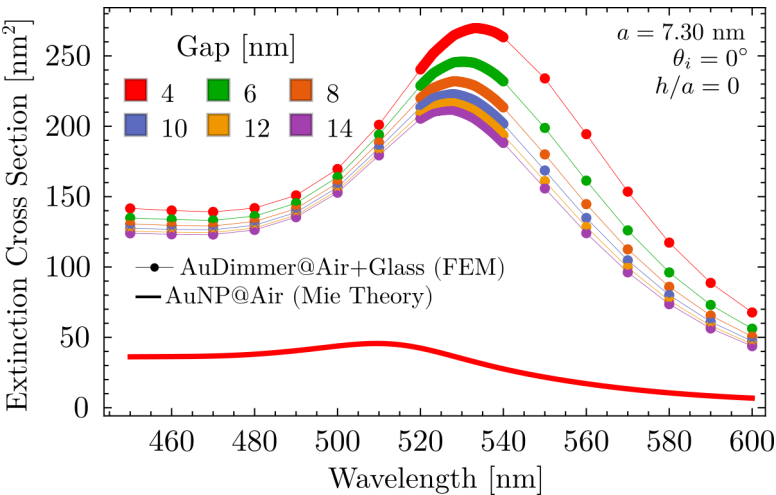
$$V_{\text{above}}^{\text{semi-sphere}} = \frac{4}{3}\pi a^3 \left[\frac{1}{2} - \frac{3}{4} \left(\frac{h}{a} \right) + \frac{1}{4} \left(\frac{h}{a} \right)^3 \right] = \frac{4}{3}\pi (a_{\text{Fit}}^{\text{CSM}})^3 = V_{\text{eq}}$$

$$\eta \left(\frac{h}{a} \right) = \frac{1}{2} - \frac{3}{4} \left(\frac{h}{a} \right) + \frac{1}{4} \left(\frac{h}{a} \right)^3 - \left(\frac{a_{\text{Fit}}^{\text{CSM}}}{a} \right)^3 = 0$$



Avances en caracterización no intrusiva

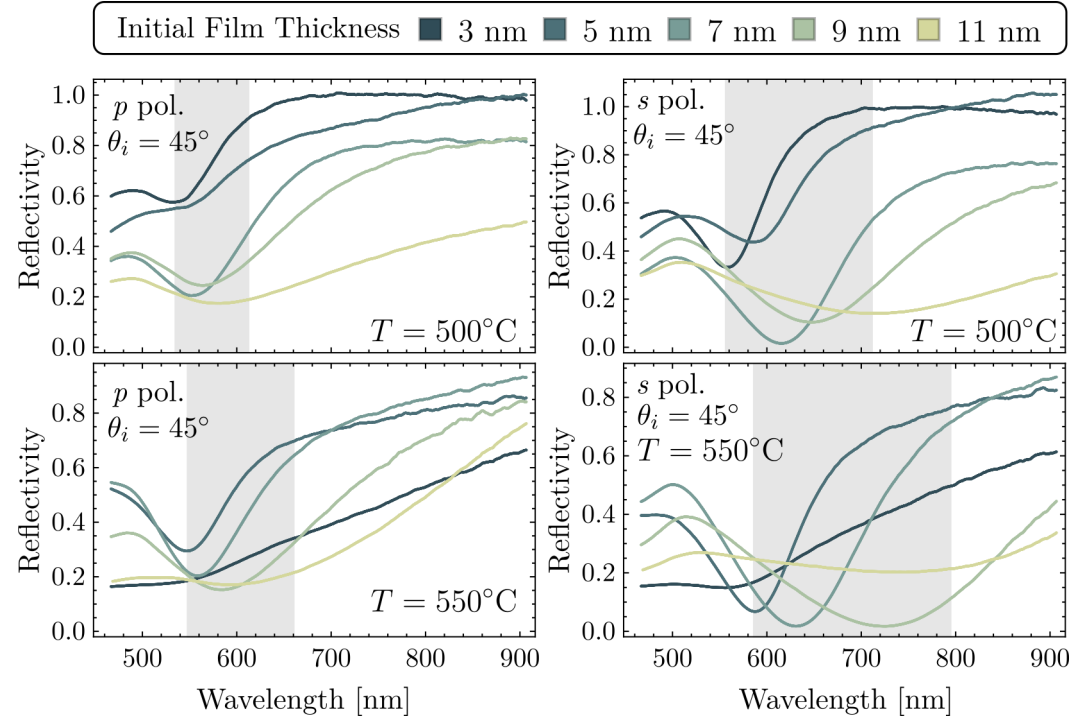
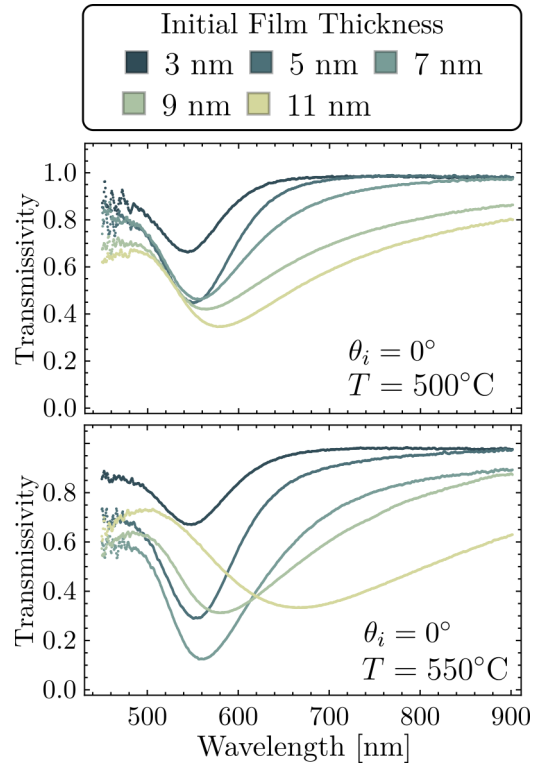
Interacción entre pares de partículas



Separación [nm]	4	6	8	10	12	14
Longitud de onda de excitación [nm]	533	53	528	528	528	527

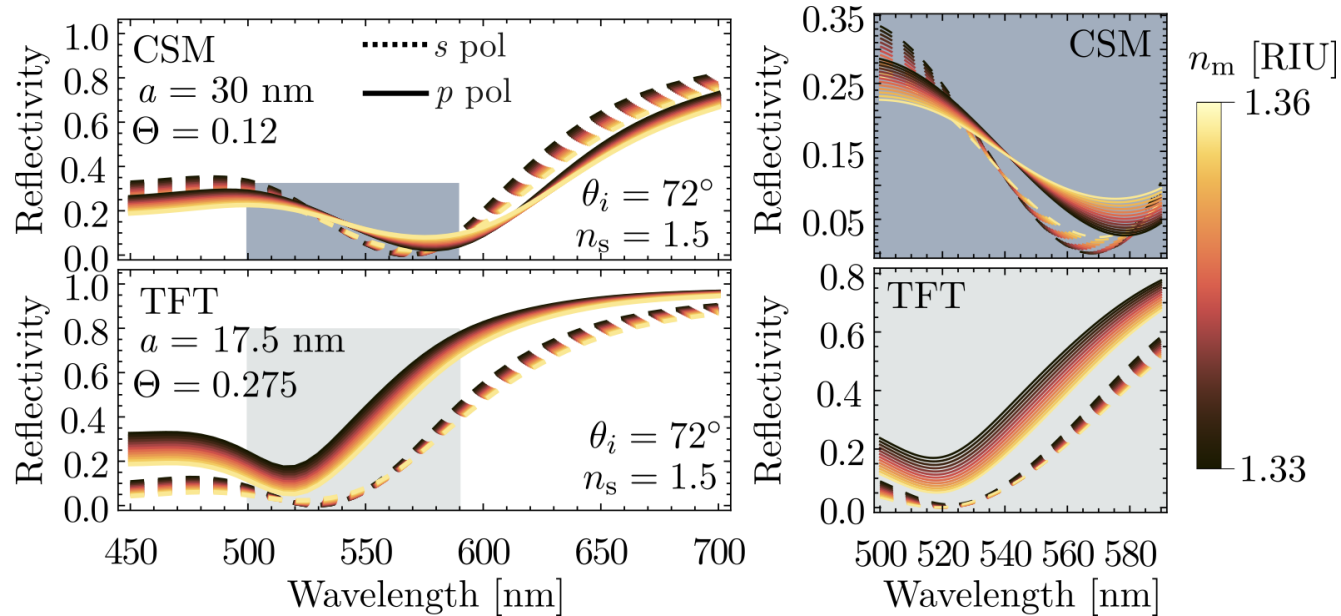
Avances en caracterización no intrusiva

Primeras mediciones de incrustamiento parcial



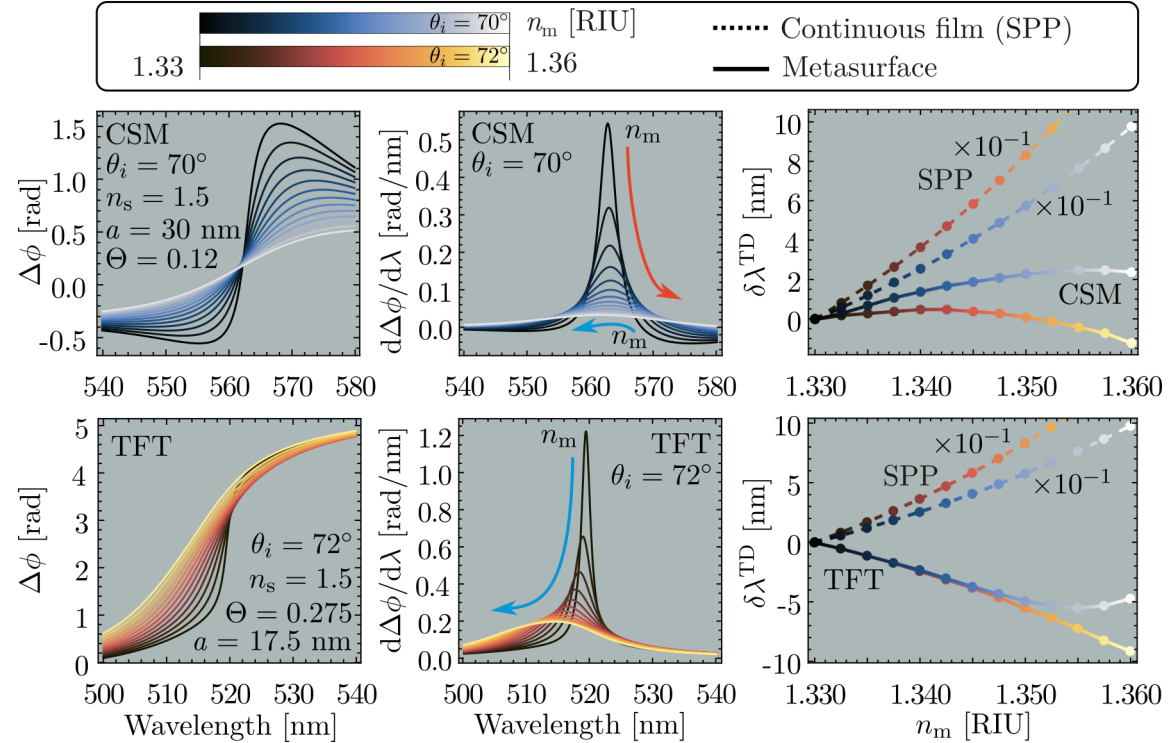
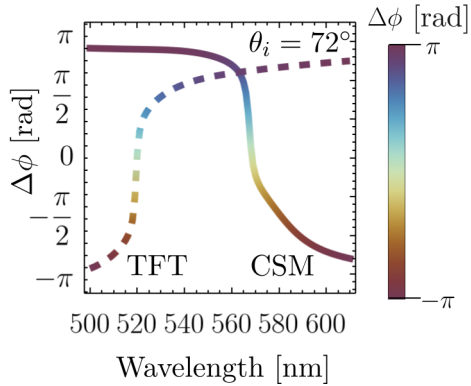
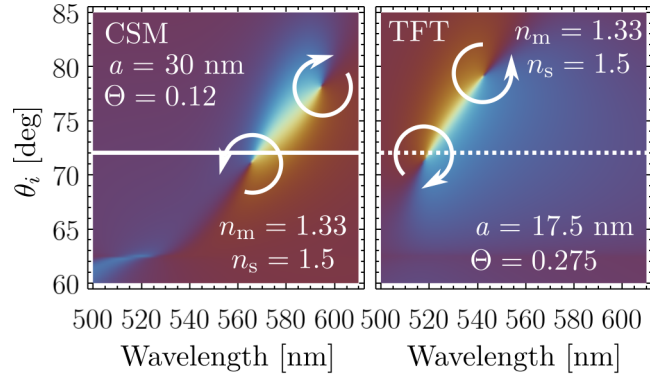
Avances en respuesta óptica y mecanismos de sensado

Esquema de medición de intensidad



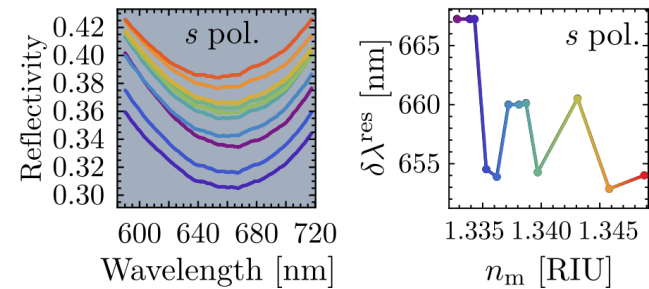
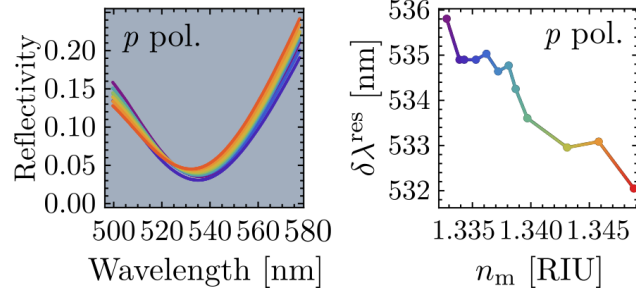
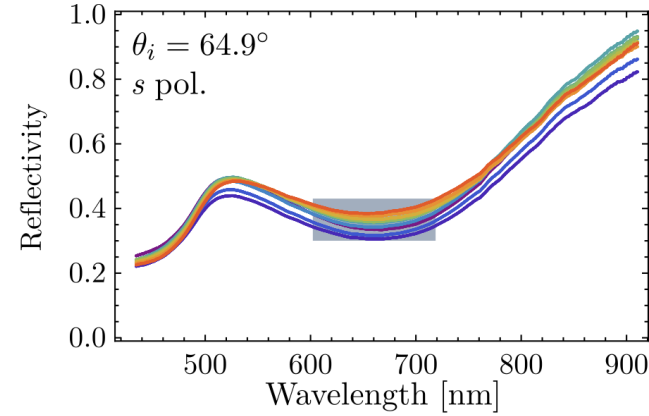
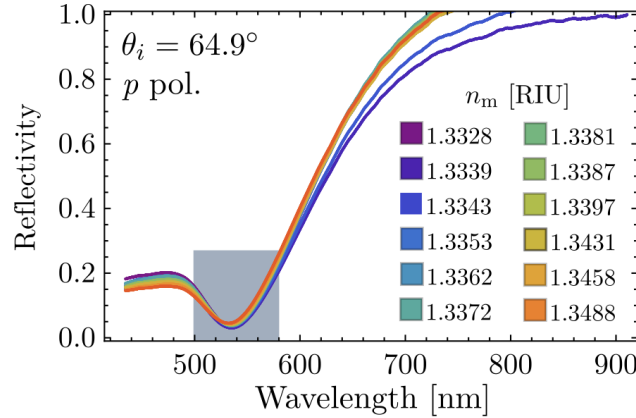
Avances en respuesta óptica y mecanismos de sensado

Esquema de medición de fase y oscuridad topológica



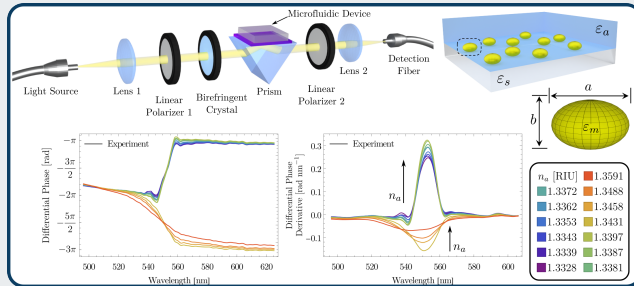
Avances en respuesta óptica y mecanismos de sensado

Evidencia experimental de corrimiento al azul



Phase Singularity in Random Gold Metasurface for Refractive Index Sensing: Experimental Demonstration and Theoretical Analysis

- ▶ J. P. Cuanalo-Fernández, N. Korneev, I. Cosme, M. B. De La Mora, **J. A. Urrutia-Anguiano**, A. Reyes-Coronado, R. Ramos-García y S. Mansurova
- ▶ J. Phys. Chem. C **128**(27): 11372–11381, 2024
- ▶ DOI: 10.1021/acs.jpcc.4c02274



Theoretical and numerical approach to substrate effects in random plasmonic metasurfaces

- ▶ **Jonathan A. Urrutia-Anguiano**, Isabel Y. Rojas-Martinez, Juan P. Cuanalo-Fernández, Alejandro Reyes-Coronado, Rubén Ramos-García y Svetlana Mansurova
- ▶ Revisión interna entre colaboradores

Spectral shift in the strong couple regime: Red- and blueshift in plasmonic disordered metasurfaces

- ▶ Juan P. Cuanalo-Fernández, **Jonathan A. Urrutia-Anguiano**, Irving Gazga-Gurrión, Nikolai Korneev, Alejandro Reyes-Coronado, Rubén Ramos-García y Svetlana Mansurova
- ▶ En proceso de escritura

Plan de trabajo 2025 - 2027

Enero - Junio 2025

- ▶ TFT con polidispersidad en tamaños
- ▶ Finalización de manuscrito
- ▶ Examen de candidatura

Julio - Diciembre 2025

- ▶ Correcciones radiativas a orden dipolar
- ▶ Homogenización para incrustamiento parcial
- ▶ Síntesis de NPs

Enero - Junio 2026

- ▶ Sustrato en CSM
- ▶ Calibración para sensado con metasuperficies ordenadas
- ▶ Escritura de tesis

Julio 2026 - Diciembre 2027

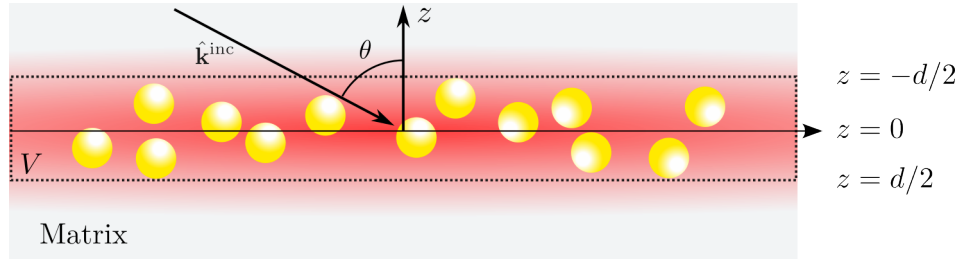
- ▶ Efectos no lineales y enfriamiento pasivo en sistemas de NPs desordenados
- ▶ Proceso de titulación en el PCF

Teoría de esparcimeinto múltiple¹

Jerarquización de Foldy-Lax

N esparcidores

- ▶ Iluminados con onda plana
- ▶ $N \rightarrow \infty$ ecuaciones
- ▶ V : Volumen semiinfinito
- ▶ $\rho(\mathbf{R})$



$$\mathbf{E}_k^{\text{exc}}(\mathbf{r}) = \mathbf{E}^{\text{inc}}(\mathbf{r}) + \sum_{\ell \neq k}^N \mathbf{E}_\ell^{\text{ind}}(\mathbf{r})$$

$$\mathbf{E}_\ell^{\text{ind}}(\mathbf{r}) = \int d^3r' \mathbb{G}(\mathbf{r}, \mathbf{r}') \times \int d^3r'' \mathbb{T}(\mathbf{r}' - \mathbf{r}_\ell, \mathbf{r}'' - \mathbf{r}_\ell) \mathbf{E}_\ell^{\text{exc}}(\mathbf{r}'')$$

$$\langle \mathbf{E}(\mathbf{r}) \rangle = \mathbf{E}^{\text{inc}}(\mathbf{r}) + \sum_{\ell=1}^N \underbrace{\left(\prod_{k=1}^N \int d^3r_k \rho(\mathbf{R}) \mathbf{E}_\ell^{\text{ind}}(\mathbf{r}) \right)}_{\langle \mathbf{E}_\ell^{\text{ind}} \rangle}$$

Aproximación de esparcidor individual (SSA)

$$\langle \mathbf{E}_\ell^{\text{ind}}(\mathbf{r}) \rangle = \int d^3r' \mathbb{G}(\mathbf{r}, \mathbf{r}') \int d^3r'' \times \int d^3r_\ell \rho(\mathbf{r}_\ell) \mathbb{T}(\mathbf{r}' - \mathbf{r}_\ell, \mathbf{r}'' - \mathbf{r}_\ell) \langle \mathbf{E}_\ell^{\text{exc}}(\mathbf{r}'', \mathbf{R}) \rangle_\ell$$

$$\langle \mathbf{E}_\ell^{\text{exc}}(\mathbf{r}'', \mathbf{R}) \rangle_\ell = \prod_{\substack{k=1 \\ k \neq \ell}}^N \int d^3r_k \rho(\mathbf{R}|\mathbf{r}_k) \mathbf{E}_\ell^{\text{exc}}(\mathbf{r}'') = \mathbf{E}^{\text{inc}}(\mathbf{r}'')$$

Aproximación Cuasicristalina (QCA)

$$\langle \mathbf{E}_\ell^{\text{exc}}(\mathbf{r}'', \mathbf{R}) \rangle_\ell = \mathbf{E}^{\text{inc}}(\mathbf{r}'') + \sum_{\substack{m=1 \\ m \neq \ell}}^N \int d^3r' \mathbb{G}(\mathbf{r}', \mathbf{r}'') \times \int d^3r''' \int d^3r_m \rho(\mathbf{r}_m) \mathbb{T}(\mathbf{r}' - \mathbf{r}_m, \mathbf{r}''' - \mathbf{r}_m) \langle \mathbf{E}_m^{\text{exc}}(\mathbf{r}''', \mathbf{R}) \rangle_{\ell, m}$$

$$\langle \mathbf{E}_m^{\text{exc}}(\mathbf{r}'', \mathbf{R}) \rangle_{\ell, m} = \prod_{\substack{n=1 \\ n \neq \ell, m}}^N \int d^3r_n \rho(\mathbf{R}|\mathbf{r}_n) \mathbf{E}_m^{\text{exc}}(\mathbf{r}'') = \langle \mathbf{E}_\ell^{\text{exc}}(\mathbf{r}'', \mathbf{R}) \rangle_\ell$$

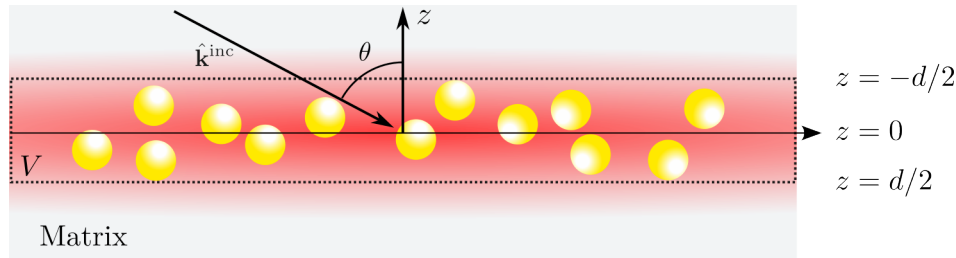
¹A. García-Valenzuela et al. JOSA A, **29**(6):1161-1179, 2012

Modelo de esparcimiento coherente (CSM)

Caso monodisperso¹

N esparcidores

- QCA
- Película de grosor $d \rightarrow 0$
- Correlación de esfera dura
- Esferas de radio a
- $\rho(\mathbf{r}_\ell) = 1/V$
- $\alpha = 2\Theta/[(ka)^2 \cos \theta]$



$$\langle \mathbf{E}_{\text{SSA}}^{\text{ind}}(\mathbf{r}) \rangle = \begin{cases} -\alpha S_i(\pi - 2\theta) \exp(i\mathbf{k}_{\text{coh}}^r \cdot \mathbf{r}) \mathbf{E}^{\text{inc}}, & z > 0 \\ -\alpha S(0) \exp(i\mathbf{k}_{\text{coh}}^t \cdot \mathbf{r}) \mathbf{E}^{\text{inc}}, & z < 0 \end{cases}$$

$$\begin{aligned} \langle \mathbf{E}_\ell^{\text{exc}}(\mathbf{r}) \rangle_\ell &= \mathbf{E}_r^{\text{exc}}(\mathbf{r}) \exp(i\mathbf{k}_{\text{coh}}^r \cdot \mathbf{r}) + \mathbf{E}_t^{\text{exc}}(\mathbf{r}) \exp(i\mathbf{k}_{\text{coh}}^t \cdot \mathbf{r}) \\ \langle \mathbf{E}_{\text{QCA}}^{\text{ind}}(\mathbf{r}) \rangle &= \begin{cases} -\alpha \left(S_j(0) \|\mathbf{E}_r^{\text{exc}}(\mathbf{r})\| + S(\pi - 2\theta) \|\mathbf{E}_t^{\text{exc}}(\mathbf{r})\| \right) \exp(i\mathbf{k}_{\text{coh}}^r \cdot \mathbf{r}) \hat{\mathbf{E}}^{\text{inc}}, & z > 0 \\ -\alpha \left(S_j(\pi - 2\theta) \|\mathbf{E}_r^{\text{exc}}(\mathbf{r})\| + S(0) \|\mathbf{E}_t^{\text{exc}}(\mathbf{r})\| \right) \exp(i\mathbf{k}_{\text{coh}}^t \cdot \mathbf{r}) \hat{\mathbf{E}}^{\text{inc}}, & z < 0 \end{cases} \\ \mathbf{E}_r^{\text{exc}}(\mathbf{r}) &= -\frac{\alpha}{2} \left(S_j(0) \|\mathbf{E}_r^{\text{exc}}(\mathbf{r})\| + S(\pi - 2\theta) \|\mathbf{E}_t^{\text{exc}}(\mathbf{r})\| \right) \hat{\mathbf{E}}^{\text{inc}}, \\ \mathbf{E}_t^{\text{exc}}(\mathbf{r}) &= \mathbf{E}^{\text{inc}}(\mathbf{r}) - \frac{\alpha}{2} \left(S_j(\pi - 2\theta) \|\mathbf{E}_r^{\text{exc}}(\mathbf{r})\| + S(0) \|\mathbf{E}_t^{\text{exc}}(\mathbf{r})\| \right) \hat{\mathbf{E}}^{\text{inc}}. \end{aligned}$$

Coeficiente de reflexión y transmisión

$$\begin{aligned} r_{\text{CSM}}^{(j)}(\theta) &= \frac{-\alpha S_j(\pi - 2\theta)}{1 + \alpha S(0) + \frac{1}{4}\alpha^2 [S^2(0) - S_j^2(\pi - 2\theta)]} \\ t_{\text{CSM}}^{(j)}(\theta) &= \frac{1 - \frac{1}{4}\alpha^2 [S^2(0) - S_j^2(\pi - 2\theta)]}{1 + \alpha S(0) + \frac{1}{4}\alpha^2 [S^2(0) - S_j^2(\pi - 2\theta)]} \end{aligned}$$

Múltiples reflexiones

Presencia de sustrato (Reflexión)

$$r^{(j)}(\theta_i) = \frac{r_{\text{sm}}^{(j)}(\theta_i) + r_{\text{CSM}}^{(j)}(\theta_t)}{1 + r_{\text{sm}}^{(j)}(\theta_i) r_{\text{CSM}}^{(j)}(\theta_t)}$$

¹A. García-Valenzuela et al. JOSA A, **29**(6):1161-1179, 2012

Teoría de película delgada (TFT)

Condiciones de frontera modificadas¹

Ecuaciones de Maxwell para campos de exceso

Límite de película delgada
Transformada de Fourier en z
 $k_z = 0$

$$\mathbf{P}^{\text{film}}(\mathbf{r}_{\parallel}) = \mathbf{D}_{\parallel}^{\text{film}}(\mathbf{r}_{\parallel}) - \varepsilon_0 E_{\perp}^{\text{film}}(\mathbf{r}_{\parallel}) \hat{\mathbf{e}}_{\perp}$$

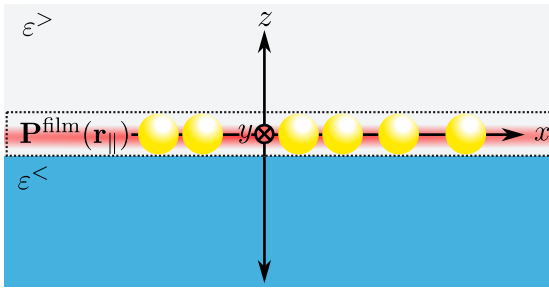
$$\mathbf{M}^{\text{film}}(\mathbf{r}_{\parallel}) = \frac{1}{\mu_0} \mathbf{B}_{\parallel}^{\text{film}}(\mathbf{r}_{\parallel}) - H_{\perp}^{\text{film}}(\mathbf{r}_{\parallel}) \hat{\mathbf{e}}_{\perp}$$

$$\langle a^{\text{film}}(\mathbf{r}_{\parallel}) \rangle = (1/2)[a^{>}(\mathbf{r}_{\parallel}) + a^{<}(\mathbf{r}_{\parallel})]$$

Ecuaciones constitutivas

$$\mathbf{P}^{\text{film}}(\mathbf{r}_{\parallel}) = \gamma \langle \mathbf{E}_{\parallel}^{\text{film}}(\mathbf{r}_{\parallel}) \rangle + \beta \langle D_{\perp}^{\text{film}}(\mathbf{r}_{\parallel}) \rangle \hat{\mathbf{e}}_{\perp}$$

$$\mathbf{M}^{\text{film}}(\mathbf{r}_{\parallel}) = \mathbf{0}$$



$$\gamma = \frac{2}{3} \frac{\Theta}{V_{\text{sp}}} \left(\frac{\alpha_{\text{sp}}}{1 + A\alpha_{\text{sp}}} \right)$$

$$\beta = \frac{2}{3} \frac{\Theta}{\varepsilon_i^2 V_{\text{sp}}} \left(\frac{\alpha_{\text{sp}}}{1 + 2A\alpha_{\text{sp}}} \right)$$

$$\alpha_{\text{sp}} = 3\varepsilon_t V_{\text{sp}} \left(\frac{\varepsilon_{\text{sp}} - \varepsilon_t}{\varepsilon_{\text{sp}} + 2\varepsilon_t} \right)$$

$$A = \frac{\varepsilon_i - \varepsilon_t}{\varepsilon_i + \varepsilon_t}$$

Condiciones de frontera

$$[D_{\perp}^{>}(\mathbf{r}_{\parallel}) - D_{\perp}^{<}(\mathbf{r}_{\parallel})] = -\nabla_{\parallel} \cdot \mathbf{P}_{\parallel}^{\text{film}}(\mathbf{r}_{\parallel})$$

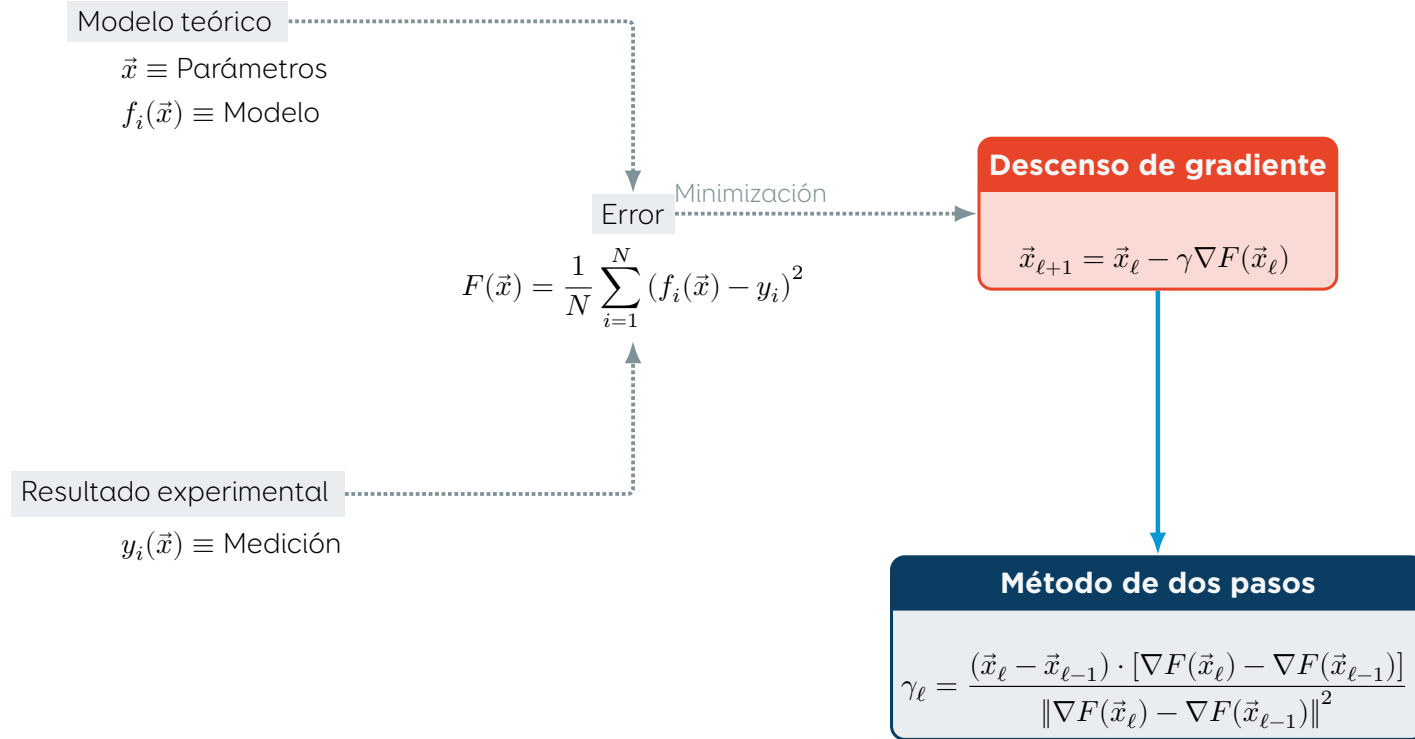
$$[B_{\perp}^{>}(\mathbf{r}_{\parallel}) - B_{\perp}^{<}(\mathbf{r}_{\parallel})] = -\nabla_{\parallel} \cdot \mathbf{M}_{\parallel}^{\text{film}}(\mathbf{r}_{\parallel})$$

$$[\mathbf{E}_{\parallel}^{>}(\mathbf{r}_{\parallel}) - \mathbf{E}_{\parallel}^{<}(\mathbf{r}_{\parallel})] = -i\omega \hat{\mathbf{e}}_{\perp} \times \mathbf{M}_{\parallel}^{\text{film}}(\mathbf{r}_{\parallel}) - \nabla_{\parallel} P_{\perp}^{\text{film}}(\mathbf{r}_{\parallel})$$

$$[\mathbf{H}_{\parallel}^{>}(\mathbf{r}_{\parallel}) - \mathbf{H}_{\parallel}^{<}(\mathbf{r}_{\parallel})] = +i\omega \hat{\mathbf{e}}_{\perp} \times \mathbf{P}_{\parallel}^{\text{film}}(\mathbf{r}_{\parallel}) - \nabla_{\parallel} M_{\perp}^{\text{film}}(\mathbf{r}_{\parallel})$$

Ajuste de parámetros

Minimización del error mediante descenso de gradiente



Metasuperficie vs. partícula individual

Longitudes de onda de excitación de una metasuperficie plasmónica en ATR

		Radio considerado [nm]	Fracción de cubierta considerada Θ	Longitud de onda de excitación [nm]	
				$\theta_i = 64.9^\circ$	$\theta_i = 72.9^\circ$
Polarización s	Experimental	5.78	0.20	536.71	552.36
		7.30	0.22	536.07	569.53
	CSM (Valores nominales)	5.78	0.20	525.05	531.53
		7.30	0.22	525.05	534.77
	CSM (Fit @ 64.9°)	3.56	0.25	525.05	-
		5.29	0.27	525.05	-
	CSM (Fit @ 72.9°)	3.45	0.25	-	531.53
		5.45	0.27	-	534.77
Polarización p	Experimental	5.78	0.20	513.44	525.69
		7.30	0.22	511.68	522.89
	CSM (Valores nominales)	5.78	0.2	525.05	534.77
		7.30	0.22	525.05	538.06
	CSM (Fit @ 64.9°)	3.29	0.17	525.05	-
		4.03	0.19	525.05	-
	CSM (Fit @ 72.9°)	3.34	0.17	-	531.53
		4.11	0.19	-	534.77